Assignment 1: Parking Fee Calculator

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Environment. VS Code 1.130.0 on Ubuntu 24.04.5 (WSL2). Verified on Windows with the Visual Studio 2022 v143 toolset, MSVC 19.44.35229, C17. PPY 0.7.0 generates standalone C; the shared `csed103` tool builds, tests, and captures the example sessions.

Design

Problem 1. A loop processes all N cars. `parking_fee` returns 0 for up to 30 minutes; otherwise, `(minutes - 30 + 9) / 10` rounds the excess up to 10-minute units. Each unit costs 500 won, capped at 5000 won. `total_fee` accumulates fees; `free_cars` counts zero-fee cars.

Problem 2. A loop checks every minute from 0 through 1440. Whenever `parking_fee(minute) <= budget`, it updates `maximum_time`. The final value is the longest affordable time, including equality.

Results and verification

Problem 1: Input N = 5 and times 0, 30, 31, 41, 131. Expected fees: 0, 0, 500, 1000, 5000 won; total 6500 won and 2 free cars. The recorded output below matches these results.

```
> ./problem1.exe
===Parking Fee Calculator===
Number of cars: 5
Car 1 parking time (minutes): 0
Car 1 fee: 0 won
Car 2 parking time (minutes): 30
Car 2 fee: 0 won
Car 3 parking time (minutes): 31
Car 3 fee: 500 won
Car 4 parking time (minutes): 41
Car 4 fee: 1000 won
Car 5 parking time (minutes): 131
Car 5 fee: 5000 won
Total fee: 6500 won, Free cars: 2
```

Problem 2: Input budget = 750 won. Expected maximum: 40 minutes (500 won); 41 minutes costs 1000 won. The recorded output matches.

```
> ./problem2.exe
===Maximum Parking Time===
Budget (won): 750
Maximum parking time: 40 minutes
```

All **44 tests passed** on MSVC (Problem 1: 9; Problem 2: 35), with no compiler warnings. Tests cover billing boundaries, the fee cap, 20 cars, non-multiple budgets, and INT_MAX. Budgets 0, 4999, and 5000 correctly allow 30, 120, and 1440 minutes. Captures are automatically rendered from actual pseudoconsole sessions, including echoed example input. `_CRT_SECURE_NO_WARNINGS` enables `scanf` without Visual Studio's deprecation diagnostic.